Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2023-P2-Model)** **Date: 04-02-2026**

Time: 02.00Pm to 05.00Pm **GTA-14** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**D**	2	**B**	3	**C**	4	**D**	5	**ABC**	6	**BCD**
7	**ABCD**	8	**3**	9	**5**	10	**0**	11	**4**	12	**2026**
13	**21**	14	**76.25**	15	**24.50**	16	**0.75**	17	**14**		

PHYSICS

18	**D**	19	**B**	20	**D**	21	**C**	22	**ABD**	23	**AD**
24	**ABCD**	25	**4**	26	**8**	27	**2**	28	**1**	29	**8**
30	**4**	31	**4.33**	32	**2166 - 2167**	33	**4**	34	**0**		

CHEMISTRY

35	**B**	36	**C**	37	**C**	38	**B**	39	**AD**	40	**ABD**
41	**ABCD**	42	**35**	43	**24**	44	**9**	45	**18**	46	**795**
47	**41**	48	**13.80**	49	**62**	50	**17**	51	**118**		

For More Join: @IITJEE_Advanced

SOLUTIONS

MATHEMATICS

1. $\Delta EFM - \Delta AFD$

$$\Rightarrow \frac{area\ \Delta EFM}{area\ \Delta AFD} = \left(\frac{MF}{FD}\right)^2$$

$$\Rightarrow FD = 2MF \quad \ldots. (1)$$

$$\Rightarrow BF = 2FD$$

Let $BD = a$ then area $\Delta EFD = \frac{1}{2} \times \frac{a}{3} \times a\frac{\sqrt{3}}{2} = \frac{a^2}{4\sqrt{3}}$

Area $\Delta AFC = \frac{1}{2} \times \left(\frac{4a}{3}\right) \times 2a\frac{\sqrt{3}}{2} = \frac{2a^2}{\sqrt{3}}$

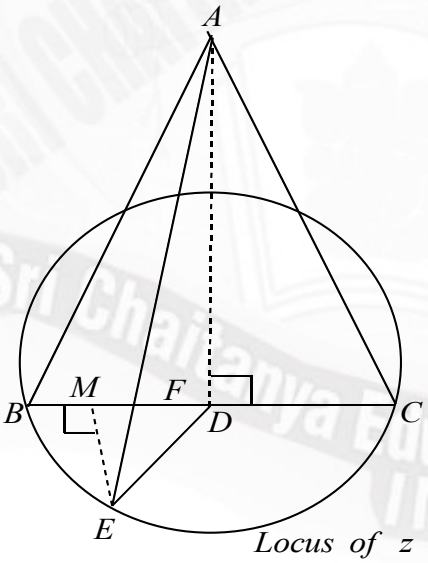
So $\frac{area\ \Delta FED}{area\ \Delta AFC} = \frac{1}{8} \quad \ldots\ldots (2)$

Now $|z - z_a|_{\max} = a\left(1+\sqrt{3}\right)$

$|z - z_a|_{\min} = a\left(\sqrt{3}-1\right)$

$$\Rightarrow \left(\frac{|z - z_a|_{\max}}{|z - z_a|_{\min}} - 2\right)^2 = 3 \quad \ldots\ldots.. (3)$$

From (2) & (3) $k = 24$



2. $\left|\overrightarrow{PQ}\right| = \sqrt{r^2 + 1 - 2r\cos\frac{\pi}{3}} = \sqrt{r^2 + 1 - r}$

& $\left|\overrightarrow{AT}\right| = \sqrt{r^2 + 1 - 2r\cos\theta}$

Where 'θ' is the angle between $\overrightarrow{OA}$ and $\overrightarrow{OT}$.

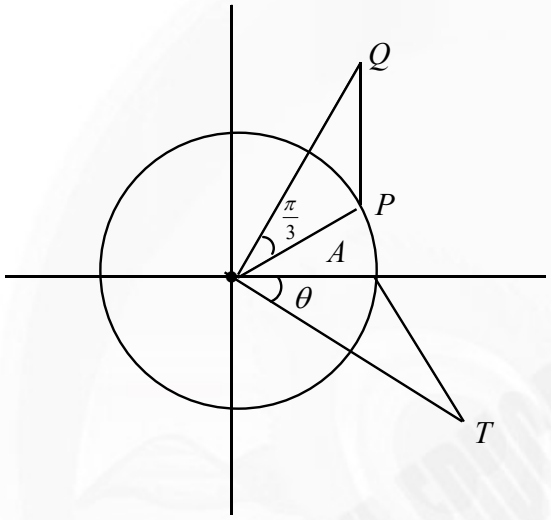
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now, $\left|\overrightarrow{PQ}\right|^2 = 4\left|\overrightarrow{AT}\right|^2$

$\Rightarrow\left(1+r^2-r\right)=4\left(r^2+1-2r\cos\theta\right)$

$\cos\theta=\dfrac{3r^2+r+3}{8r}\Rightarrow -1\le\dfrac{3r^2+r+3}{8r}\le 1;\quad r>0$

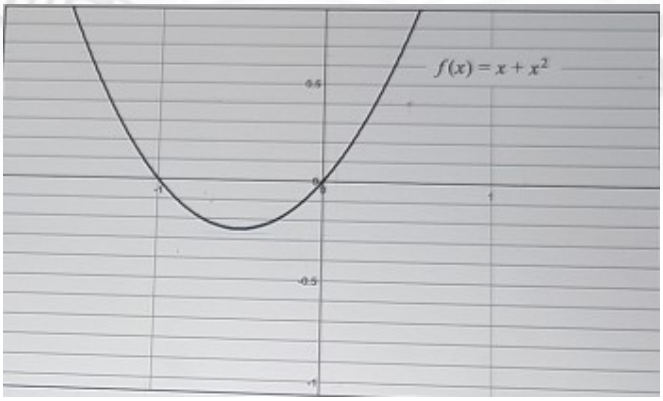
$\Rightarrow r\in\left[\dfrac{7-\sqrt{3}}{6},\dfrac{7+\sqrt{3}}{6}\right]$



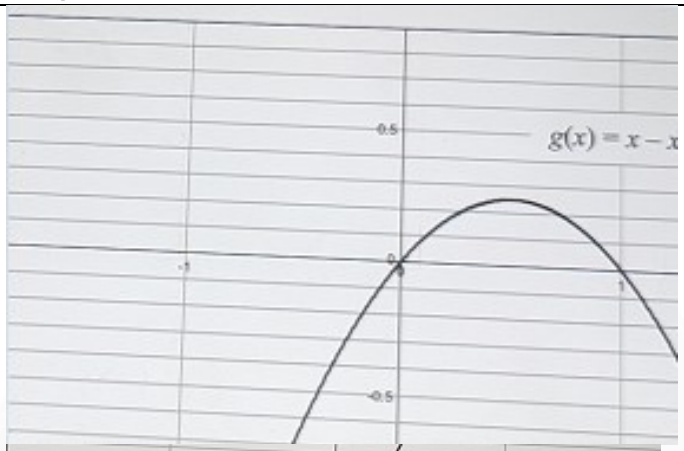
3. 1) is False. Look at $f(x)=\lfloor x\rfloor, g(x)=|x|$. At $x=0$ $y=f.g$ is not differentiable. The statement

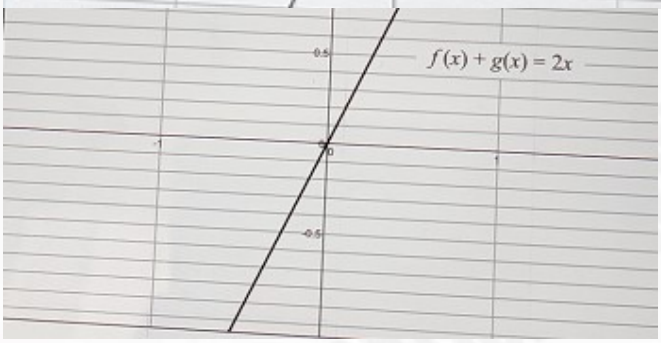
3) is false. Look at $f(x)=\lfloor x\rfloor, g(x)=\begin{cases}0 & x\neq 0\\ 1 & x=0\end{cases}$. We see f is not differentiable at $x=0$ and $g(0)\neq 0$ but $y=f\cdot g$ is differentiable at $x=0$. Statement (5) is incorrect. Look at $f(x)=\lfloor x\rfloor, g(x)=x$. At $x=0$ these two satisfies the hypothesis but $y=f\cdot g$ is not differentiable at $x=0$.

8) Both functions $f(x)=x+x^2$ and $g(x)=x-x^2$ are not monotone on $\mathbb{R}$, but their sum $f(x)+g(x)=2x$ is monotone on $\mathbb{R}$.



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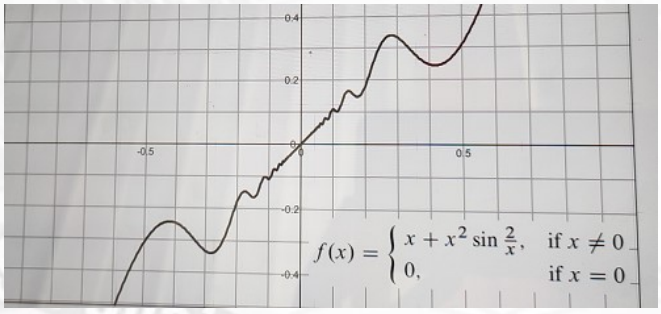




9)

$$f(x)=\begin{cases} x+x^2\sin\dfrac{2}{x}, & if\ x\neq 0 \\ 0, & if\ x=0 \end{cases}$$

Is increasing at the point $x=0,$ but it is not increasing in any neighbourhood $(-\delta,\delta)$, where $\delta>0.$



Comment: The definition of a function increasing at a point is: A function $f(x)$ is said to be increasing at the point $x=a$ if in certain neighbourhood $(x-\delta,x+\delta),\ \delta>0$ the following is true.

If $x<a$ then $f(x)<f(a)$ if $x>a$ then $f(x)>f(a).$

4. $AA^T=A^TA=I,\ \ BB^T=B^TB=I$

Now $\left(ABA^{-1}\right)^T\left(ABA^{-1}\right)=\left(A^T\right)^{-1}B^TA^TABA^{-1}$

$=\left(A^T\right)^{-1}A^{-1}=\left(AA^T\right)^{-1}=I$

$\Rightarrow\left(ABA^{-1}\right)^T\left(ABA^{-1}\right)=I$

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$\because \quad \left(ABA^{-1}\right)^T = ABA^{-1} \left(\because \ symmetric\right)$

$\Rightarrow \left(ABA^{-1}\right)^2 = I \Rightarrow ABA^{-1}$ is involutory matrix and $\left(ABA^{-1}\right)^{2017} = ABA^{-1}$

$\because \quad ABA^{-1} \neq I, -I$

$\Rightarrow ABA^{-1} = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$, where $\alpha^2 + \beta\gamma = 1$

$\Rightarrow tr\left(ABA^{-1}\right) = 0$

$\because \left(ABA^{-1}\right)^2 - I \Rightarrow ABA^{-1}ABA^{-1} = I$

$\Rightarrow B^2 = I \Rightarrow tr\left(B^2\right) = 2$

and $B = I, -I$

$\Rightarrow B = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}, \alpha^2 + \beta\gamma = 1 \Rightarrow |B| = -1$

$\left(ad\frac{B}{\sqrt{2}}\right)^2 = \frac{1}{2}\left(adjB\right)^2 = \frac{1}{2}adjB^2 = \frac{1}{2}$

5. α

Where Im = imaginary part.

$= \text{Im}\left[\sum_{r=0}^{100} {}^{100}C_r . \left(e^{ix}\right)^r\right]$

$= \text{Im}\left[\left(1 + e^{ix}\right)^{100}\right]$

$= \text{Im}\left(1 + \cos x + i\sin x\right)^{100}$

$= \text{Im}\left(2\cos^2\frac{x}{2} + 2i\sin\frac{x}{2} \times \cos\frac{x}{2}\right)^{100}$

$= \text{Im}\left(2\cos\frac{x}{2}\left(\cos\frac{x}{2} + i\sin\frac{x}{2}\right)\right)^{100}$

$= 2^{100}.\cos^{100}\frac{x}{2}.\sin\left(50x\right)$

$\sum_{r=0}^{50} {}^{50}C_r\, a^r \times b^{50-r} \times \cos\left[rB - \left(50-r\right)A\right]$

$= \text{Re}\left(\sum_{r=0}^{50} {}^{50}C_r\, a^r \times b^{50-r} \times e^{i\left[rB-(50-r)A\right]}\right)$

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$$= \text{Re}\left[\sum_{r=0}^{50} {}^{50}C_r \left(a \times e^{iB}\right)^r \times \left(b \times e^{-iA}\right)^{50-r}\right]$$

$$= \text{Re}\left(ae^{iB} + be^{-iA}\right)^{50}$$

$$= \text{Re}\left(a\cos B + ia\sin B + b\cos A - ia\sin A\right)^{50}$$

$$= \text{Re}\left(a\cos B + b\cos A\right)^{50}$$

$$\left[\because\ a\sin B = b\sin A \ \ because\ \ \frac{a}{\sin A} = \frac{b}{\sin B}\right]$$

$$= c^{50} \qquad \left[\because\ a = b\cos C + c\cos B,\ b = c\cos A + a\cos C\ \ and\ \ c = a\cos B + b\cos A\right]$$

$$= \left(a + b + c - a - b\right)^{50}$$

$$= \left(25 - a - b\right)^{50}$$

$$= \frac{\sum_{r=0}^{50} {}^{50}C_r . \sin^2 rx}{\sum_{r=0}^{50} {}^{50}C_r . \cos^2 rx}$$

$$= \frac{\sum_{r=0}^{50} {}^{50}C_{50-r} . \sin 2(50-r)x}{\sum_{r=0}^{50} {}^{50}C_{50-r} . \cos 2(50-r)x}$$

$$= \frac{\sum_{r=0}^{50} {}^{50}C_r \left[\sin 2rx + \sin 2(50-r)x\right]}{\sum_{r=0}^{50} {}^{50}C_r \left[\cos 2rx + \cos 2(50-r)x\right]}$$

$$= \frac{\sum_{r=0}^{50} {}^{50}C_r . 2\sin(50x).\cos(2r-50)x}{\sum_{r=0}^{50} {}^{50}C_r . \sin(50x).\cos(2r-50)x}$$

$$= \tan(50x)$$

$$\Rightarrow f(\pi/8) = \tan\left(50 \times \frac{\pi}{8}\right)$$

$$= \tan\left(\frac{25\pi}{4}\right)$$

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$$= \tan\left(6\pi + \frac{\pi}{4}\right) = 1$$

6. $$f'(x) = \begin{vmatrix} 1 & 1 & ..... & 1 \\ x & x+a_2 & ..... & x \\ \vdots & \vdots & & \vdots \\ x & x & ..... & x+a_n \end{vmatrix} + \begin{vmatrix} x+a_1 & x & ... & x \\ 1 & 1 & ... & 1 \\ \vdots & \vdots & & \vdots \\ x & x & ... & x+a_n \end{vmatrix} + \begin{vmatrix} x+a_1 & x & ... & x \\ x & x+a_2 & ... & x \\ \vdots & \vdots & & \vdots \\ 1 & 1 & ... & 1 \end{vmatrix}$$

Its first derivative is

$$f'(x) = \begin{vmatrix} 1 & 1 & ..... & 1 \\ x & x+a_2 & ..... & x \\ \vdots & \vdots & \ddots & \vdots \\ x & x & ..... & x+a_n \end{vmatrix} + \begin{vmatrix} x+a_1 & x & ... & x \\ 1 & 1 & ... & 1 \\ \vdots & \vdots & \ddots & \vdots \\ x & x & ... & x+a_n \end{vmatrix} + ... + \begin{vmatrix} x+a_1 & x & ... & x \\ x & x+a_2 & ... & x \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & ... & 1 \end{vmatrix}.$$

Proceeding one step further, we see that the second derivative derivative of f consists of two types of determinants: some that have a row of 0's, and others that have two rows of 1's. In both cases the determinants are equal to zero, showing that $f''(x) = 0$. It follows that f itself must be a linear function,

$$f(x) = f(0) + f'(0)x.$$

One finds immediately that $f(0) = a_1 a_2 .... a_n$. To compute

$$f'(0) = \begin{vmatrix} 1 & 1 & ..... & 1 \\ 0 & a_2 & ..... & 0 \\ \vdots & \vdots & \ddots & \vdots \\ x & x & ..... & a_n \end{vmatrix} + \begin{vmatrix} a_1 & 0 & ... & 0 \\ 1 & 1 & ... & 1 \\ \vdots & \vdots & \ddots & \vdots \\ x & x & ... & a_n \end{vmatrix} + ... + \begin{vmatrix} a_1 & 0 & ... & 0 \\ x & a_2 & ... & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & ... & 1 \end{vmatrix}$$

Expand each determinant along the row of 1's. The answer is

$$f'(0) = a_2 a_3 .... a_n + a_1 a_3 .... a_n + ... + a_1 a_2 .... a_{n-1},$$

Whence

$$f(x) = a_1 a_2 .... a_n \left[1 + \left(\frac{1}{a_1} + \frac{1}{a_2} + ... + \frac{1}{a_n}\right)x\right].$$

Substituting $x = 1$, we obtain the formula from the statement.

7. We have,

Value	4	5	6	8	9	11	12
Frequency	5	f_1	1	f_2	2	3	1

Sum of frequencies is 19.

$\therefore \quad 5 + f_1 + 1 + f_2 + 2 + 3 + 1 = 19$

or $\quad f_1 + f_2 = 7 \qquad$ --- (1)

Also, median is 6

So, cumulative frequency till 6 must be more than 9.5.

$\therefore \quad 5 + f_1 + 1 = 4$

So, $f_2 = 3$

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$\therefore$ Mean, $\bar{x} = \dfrac{\sum x_i f_i}{\sum f_i} = \dfrac{20+20+6+24+18+33+12}{19}$

$= \dfrac{133}{19} = 7$

So, Mean Deviation about mean,

$$\alpha = \frac{\sum f_i \left| x_i - \bar{x} \right|}{\sum f_i} = \frac{15+8+1+3+4+12+5}{19} = \frac{48}{19}$$

Also, mean deviation about median,

$$\beta = \frac{\sum f_i \left| x_i - 6 \right|}{\sum f_i} = \frac{10+4+0+6+6+15+6}{19} = \frac{47}{19}$$

And, variance,

$$\sigma^2 = \frac{\sum f_i \left(x_i - \bar{x} \right)^2}{\sum f_i} = \frac{45+16+1+3+8+48+25}{19} = \frac{146}{19}$$

8. The given equation can be re-written as quadratic in y

$$y^2 + \left(8x - 2x^2\right) y + \left(3x^2 + 6x\right) = 0$$

Its discriminant to equal to

$$\Delta = \left(64x^2 - 32x^3 + 4x^4\right) - 4\left(3x^2 + 6x\right) = 4x\left(x^3 - 8x^2 + 13x - 6\right)$$

$$\Rightarrow = 4x(x-6)(x-1)^2$$

For solution in integers this discriminant should be perfect square this happens if and only if

$$z^2 = x(x-6) = (x-3)^2 - 9$$

$$9 = (x-3)^2 - z^2 = (x+z-3)(x-z-3)$$

For some integer z

Checking all factorisation

$$9 = (-9) \times (-1) = (-3) \times (-3) = (-1) \times (-9) = 9 \times 1 = 3 \times 3 = 1 \times 9$$

This gives all possible pairs (x, y) as

$$(x, y) = (-2, 0), (-2, 24), (0, 0), (8, 4), (8, 60), (6, 12)$$

Therefore, exactly three pairs of natural number (x, y) are possible

9. The given expression can be re written as

$$\sum_{x \in (p, q, r, s)} \left(2x^2 + (1-x)^2\right) \text{ where } 0 \le x \le 1$$

Consider the function $f(x) = 2x^2 + (1-x)^2, \ 0 \le x \le 1$

So, $f'(x) = 4x - 2(1-x)$

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For critical points $f'(x)=0$ or $x=1/3$

Thus, set of critical points is $\left\{0, \frac{1}{3}, 1\right\}$

Now $f(0)=1,\ f\left(\frac{1}{3}\right)=\frac{2}{3},\ f(1)=2$

Thus, maximum value of $f(x)$ is 2 and minimum value of $f(x)$ is 2/3.

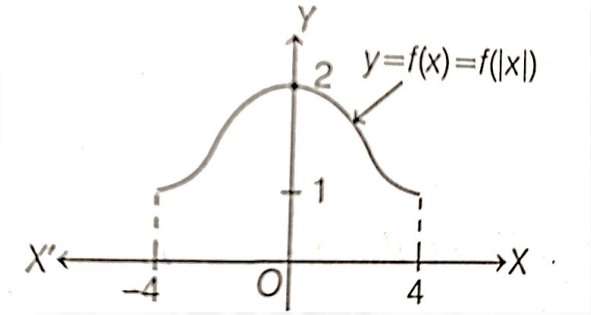
Therefore, maximum value of given expression is $4\times 2=8$ and minimum value is $4\times(2/3)=\frac{8}{3}$

10. For $\Phi(x)=f(|x|)$

$-3\le|x|\le 4$

$\Rightarrow 0\le|x|\le 4$

$\Rightarrow -4\le x\le 4$



$\therefore$ Domain of $\Phi(x)=[-4, 4]$

So, $(\beta-\alpha)=8$

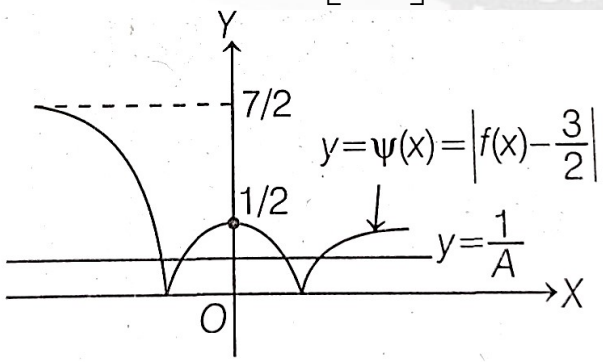
Also, range of $\Phi(x)=[1, 2]$

$\therefore \quad (\gamma+\delta)=3$

$\therefore \quad (\beta-\alpha+\gamma+\delta)=11=\lambda$

Hence, difference of unit's and ten's place digits present in $\lambda=1-1=0$ ..... (i)

Also, range of $\Psi(x)=\left[0, \frac{7}{2}\right]$ and number of solutions of $\Psi(x)=\frac{1}{4}$ is n = 4



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So, $n+p+2q=4+0+7=11=\mu$

$\therefore$ Sum of digits in the value of

$\mu=1+1=2$ ……. (ii)

Therefore, $k=\dfrac{0}{2}=0$

[From eqs. (i) and (iii)]

Hence, $\left(2\sin k+\tan 2k\right)$

$=2\times\sin 0^0+\tan 0^0=0$

11. $f(x)=\sum_{m=0}^{\infty}a_m x^{2m}(1+x)$ where $a_m=\dfrac{1.3.5.7.....(2m-1)}{2.4.6.8....(2m)}$ terms

$(2m-1)$ value or terms

$$\Rightarrow \boxed{f(x)=\sum_{m=0}^{\infty}a_m x^{2m}(1+x)} \text{ ----- (1) } m=1,\quad a_1=\frac{1}{2}\left(x^2+x^3\right)$$

$2m!=1\,.\,2\,.\,3\,.\,4------2m$

$=\{1.3.5---(2m-1)\}\ \{2.4.6---(2m)\}$

$\Rightarrow 2m!=O_m\,.\,E_m$

$and\quad E_m=2\,.\,4\,.\,6---(2m)$

$=2(1.3.5---m)$

$=2^m\ m!$

$$\left|\ O_m=\frac{2m!}{2^m m!}\ \right| \quad a_m=\frac{O_m}{E_m}=\frac{(2m)!}{4^m(m!)^2}$$

$$a_m=\frac{1}{4^m}\ {}^{2m}c_m$$

$$\sum_{m=0}^{\infty}a_m x^{2m}=\sum_{m=0}^{\infty}{}^{2m}c_m\left(\frac{x^2}{4}\right)^m$$

($\because$ generalized Binomial coefficient

$$\alpha_{c_k}=\frac{\alpha(\alpha-1)(\alpha-2)---(\alpha-k+1)}{k!},\ \alpha\in\mathbb{R},\ k\geq 0)$$

$$\left(\frac{-1}{2}\right)_{c_k}=\frac{\left(\frac{-1}{2}\right)\left(\frac{-1}{2}-1\right)----\left(\frac{-1}{2}-k+1\right)}{k!}$$

$$=\frac{\left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right)----\left(\frac{2k-1}{2}\right)}{k!}$$

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$$\Rightarrow \left(-1/2_{C_k}\right) = (-1)^k \frac{1.3.5---(2k-1)}{2^k.k!}$$

$$\boxed{O_k = \frac{(2k)!}{2^k.k!}};$$

$$\left(-1/2_{C_k}\right) = \frac{(-1)^k\ 1.3.5---(2k-1)}{2^x.k!} \times \frac{2.4.6---2k}{2.4.6---2k} = (-1)^k \cdot \frac{(2k)!}{2^{2k}.(k!)^2}$$

Multiplying on both sides with $(-4t)^k$

$$\sum_{m=0}^{\infty}\left(\frac{-1}{2_{c_k}}\right)(-4t)^k = \sum_{m=0}^{\infty}(-1)^k.\frac{(2k)!}{2^{2k}.(k!)^2}.(-4t)^k = \sum_{m=0}^{\infty}(-1)^k.\frac{(2k)!}{4^k.(k!)^2}.(-1)^k.4^k.t^k$$

$$= \sum_{m=0}^{\infty}\frac{(2k)!}{(k!)^2}.t^k = \sum_{m=0}^{\infty}2k_{c_k}t^4 = (1-4t)^{\frac{-1}{2}}$$

$$\therefore \sum_{m=0}^{\infty}2m_{c_m}(t)^k = \frac{1}{\sqrt{1-4t}}, |t| < \frac{1}{4}\left(\because t = \frac{x^2}{4}\right)$$

$$\Rightarrow \sum_{m=0}^{\infty}a_m x^{2m} = \frac{1}{\sqrt{1-x^2}} \qquad \therefore f(x)\frac{(1+x)}{\sqrt{1+x^2}},$$

$$f\left(\frac{15}{17}\right) = \frac{1+\frac{15}{17}}{\sqrt{1-\frac{225}{289}}} = \frac{32}{17}\times\frac{17}{8}$$

12. $$\sum_{k=1}^{n}\left[(k+3)^2-(k+2)\right](k+2)! + 3(3!) = 2029.(2029!)$$

$$\Rightarrow \sum_{k=1}^{n}\left[(k+3)(k+3)! - (k+2)(k+2)!\right] + 3(3!) = 2029.(2029!)$$

$$\Rightarrow (n+3)((n+3)!) = 2029.(2029!)$$

$$\Rightarrow n+3 = 2029$$

$$\Rightarrow n = 2029 - 3 = 2026$$

13. Given system of linear equations

$$x + y + z = 4$$

$$2x + y + 3z = 6$$

$$x + 2y + mz = n$$

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Now, $\Delta_1 = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 1 & 2 & m \end{vmatrix}$

$= 1(m-6) - 1(2m-3) + 1(4-1)$

$= m - 6 - 2m + 3 + 3 = m - 2m$

$= -m$

$\Delta_1 = \begin{vmatrix} 4 & 1 & 1 \\ 6 & 1 & 3 \\ n & 2 & m \end{vmatrix}$

$= 4(m-6) - 1(6m - 3n) + 1(12 - n)$

$= 4m - 24 - 6m + 3n + 12 - n$

$= -2(m - n + 6)$

$\Delta_1 = \begin{vmatrix} 1 & 4 & 1 \\ 2 & 6 & 3 \\ 1 & n & m \end{vmatrix} = -2m - n + 6$

$\Delta_1 = \begin{vmatrix} 1 & 1 & 4 \\ 2 & 1 & 6 \\ 1 & 2 & n \end{vmatrix} = 6 - n$

By Cramer's rule,

$x = \frac{\Delta_1}{\Delta} = \frac{-2(m - n + 6)}{-m}$

$= \frac{2(m - n + 6)}{-m}$

$y = \frac{\Delta_2}{\Delta} = \frac{2m + n - 6}{m}$

and $z = \frac{\Delta_3}{\Delta} = \frac{n - 6}{m}$

Now, for unique solution, $m \neq 0,\ n \in R$

$\therefore$ Total number of ordered pairs $(m,\ n)$

$= 20 \times 21 = 420$

i.e., $P = 420$

For no solution, $m = 0,\ n \neq 6$

$\therefore$ Total number of ordered pairs of $(m,\ n)$

$= 1 \times 20 = 20$

i.e., $Q = 20$

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For infinite many solutions, $m = 0,\ n = 6$

$= 1 \times 1 = 1$

i.e., $R = 1$

Hence, $\frac{P}{QR} = \frac{420}{20 \times 1} = 21$

14. p_1 = probability that maximum of chosen numbers is at least 81

$\Rightarrow p_1 = 1 -$ probability that maximum of chosen number is atmost 80

$\Rightarrow p_1 = 1 - \left(\frac{80}{100}\right)^3 = \frac{61}{125}$

So, $\frac{625}{4} p_1 = \frac{625}{4} \times \frac{61}{125} = \frac{305}{4}$

$\therefore \frac{625}{4} p_1 = 76.25$

15. p_2 = probability that minimum of chosen number is at most 40

$\Rightarrow p_2 = 1 -$ probability that minimum of chosen number is at least 41

$\Rightarrow p_2 = 1 - \left(\frac{60}{100}\right)^3 = \frac{98}{125}$

So, $\frac{125}{4} p_2 = \frac{125}{4} \times \frac{98}{125} = \frac{98}{4}$

$\therefore \frac{125}{4} p_2 = 24.50$

16 & 17.

$$I = \int_0^1 \sqrt{2 + 2\sqrt{2x - x^2} + \left(f'(x)\right)^2}\, dx$$

$$= \int_0^1 \sqrt{\left(\sqrt{(2-x)}\right)^2 + \left(\sqrt{x}\right)^2 + 2\sqrt{(2-x)x} + \left(f'(x)\right)^2}\, dx$$

$$= \int_0^1 \sqrt{\left(\sqrt{2-x} + \sqrt{x}\right)^2 + \left(f'(x)\right)^2}\, dx \longrightarrow \text{minimize}$$

$$\left(\because A^2 + B^2 \geq \frac{(A+B)^2}{2} \Rightarrow (A-B)^2 \geq 0 \Rightarrow A \geq B \text{ for minimize}\right)$$

$$= I \geq \frac{1}{\sqrt{2}} \int_0^1 \sqrt{\left(\sqrt{2-x} + \sqrt{x} + f'(x)\right)^2}\, dx$$

$$\geq \frac{1}{\sqrt{2}} \int_0^1 \left\{\sqrt{2-x} + \sqrt{x} + f'(x)\right\} dx$$

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$$\geq \frac{1}{\sqrt{2}}\left\{\frac{-2}{3}(2-x)^{\frac{3}{2}}+\frac{2}{3}x^{\frac{3}{2}}+f(x)\right]_0^1\}$$

Given $g(x)=\sqrt{x}$

$A \geq B \Rightarrow A-B \geq 0$

Minimized $\sqrt{2-x}+\sqrt{x}=f'(x)$

$\int \sqrt{2-x}+\sqrt{x}\ dx=\int f'(x)dx$

$f(x)=\frac{-2}{3}(2-x)^{\frac{3}{2}}+\frac{2}{3}x^{\frac{3}{2}}+C$

$f(1)=2 \Rightarrow C=2$

$f(x)=\frac{-2}{3}(2-x)^{\frac{3}{2}}+\frac{2}{3}x^{\frac{3}{2}}+2$

$f(2)=\frac{2}{3}2^{3/2}+2=\frac{6+4\sqrt{2}}{3}=\beta,\ \alpha=\frac{8}{3}$

$\therefore 3\alpha+3\beta-4\sqrt{2}=8+6=14$

$h(x)=2\sqrt{6}\,g(x)$

$y=2\sqrt{6}\,g(x)$

$y=2\sqrt{6}\sqrt{x}$

$y^2=24x,\quad xy=2$

$4a=24 \Rightarrow a=6$

$P(6t^2,12t)$ point on parabola $\overrightarrow{AB}$ is

$yy_1-2a(x+x_1)=0$

$ty=x+6t^2$

$x-ty+6t^2=0$ ---- (1)

Equation of chord AB using $M(h,k)$ as mid-point

$S_1=S_{11}$

$xk+yh-2hk=0$ ---- (2)

(1)& (2) represent same lines

$$\frac{h}{-t}=\frac{k}{1}=\frac{-2hk}{6t^2} \Rightarrow K^2=-3h$$

Locus of m: $y^2=-3x$

$a=\frac{3}{4}$, and distance $x=\frac{3}{4}\cong 0.75$

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PHYSICS

18. consider a square of side 4cm centered around the conductor

By ampere's law $\oint \bar{B}.d\bar{l} = \mu_0 i$, $\int \bar{B}.d\bar{l}$ for the given line $= \frac{\mu_0 i}{8} = \frac{4\pi \times 10^{-7} \times 2}{8} = \pi \times 10^{-7}$

19. Along length of page it completes one oscillation and along width two oscillations every time.

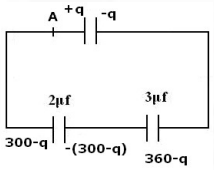
20. $\frac{1}{n}\left(\frac{dw}{dT} + \frac{du_{gas}}{dT}\right) = \frac{1}{n}\left[\frac{SdA}{dT} + \frac{du_{gas}}{dT}\right] = \frac{1}{n}\left[S\left(\frac{3nR}{2S}\right) + \frac{3}{2}nR\right]$ =3R

21. Let 'f' fraction of rope is not touching

$2T\sin\theta = fmg \quad \frac{\mu}{2}(1-f)mg\cos\theta = \frac{1}{2}(1-f)mg\sin\theta + \frac{f}{2}mg\sin\theta$

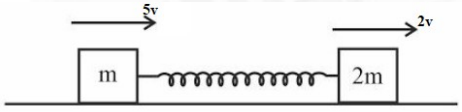
$\theta = 22.5^0, \; fraction = \frac{\sqrt{2}-1}{\sqrt{2}+1}$

22.



$V_A - \frac{q}{1.5} - \frac{360-q}{3} + \frac{300-q}{2} = V_A \quad \therefore q = 180$

23.

24. 

$V_{cm} = \frac{m(5V) + 2m(2V)}{3m} = 3V$

In the centre of mass frame, both the blocks oscillate simple harmonically.
At maximum deformation, both the blocks move with velocity of centre of mass.

$U_{s_1 \max} = K_i - K_{least} = \frac{1}{2}m\left[(5V)^2 + 2(2V)^2\right] - \frac{1}{2}(3m)(3v)^2$

$= mV^2\frac{[33-27]}{2} = 3mV^2$

Maximum speed of '2m' is observed when the spring is at natural length

$V_{B,\max} = V_{cm} + V_{B,CM|\max} = 3V + V = 4V$

$V_A, \min = \left|V_{cm} - V_A, cm|\max\right| = 3V - 2V = V$

Relative speed is independent of frame of reference $V_{rel}|\max = 3V$

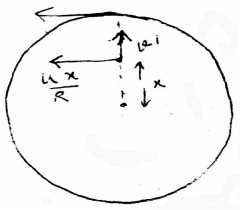
25. $V_B - V_A = -\int_A^B \bar{E}.d\bar{l} = a\left(\frac{1}{r_1} - \frac{1}{r_2}\right) = 16 \; \therefore x = 4$

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26. $C_{eq} = \frac{5}{3}.\frac{\epsilon_0 A}{d}$

27. $K_{eff} = k + k\cos^2\theta$

28. The chasers should always maintain same 'ω' as the first boy. So he has two components of velocity. $V_t = \frac{ux}{R}$ and $V_r = V'$



Given, $V_t^2 + V_l^2 = u^2$ and $V_r = \frac{dx}{dt}$ $\frac{dx}{dt} = \sqrt{u^2 - \frac{u^2x^2}{R^2}}$ $\frac{dx}{\sqrt{1-\frac{x^2}{R^2}}} = u\,dt$

Integrating $R\sin^{-1}\left(\frac{x}{R}\right)\Big|_0^R = uT \Rightarrow T = \frac{R}{u}\frac{\pi}{2}$

29. $m \times 0.5 \times 5 + 1 \times 80 = 100$ $m = \frac{20}{25} = 8$

30. $q = CV = C(Blv)$ $\frac{dq}{dt} = Blc\frac{dv}{dt}$ $-(1)$

$mg - B\left(\frac{dq}{dt}\right)l = m\frac{dv}{dt}$ $mg = \left(m + B^2l^2C\right)\frac{dv}{dt}$

$a = \frac{dv}{dt} = \frac{mg}{mB^2l^2C} = \frac{mg}{m+4m} = \frac{9}{5} = 2ms^{-1}$ $v = at = 2 \times 2 = 4ms^{-1}$

31 & 32.

$y = \left(\frac{\mu g - \mu m}{\mu m}\right)\left(\frac{tD}{d}\right) = 4.33mm$

For maxima at 0 Path difference $= n\lambda$ Or $\left(\frac{\mu_2}{\mu_1} - 1\right)t = n\lambda$ $= \frac{1.3 \times 10^{-6}}{n}m$

White light λ varies from 400 to 700 nm

For $n = 2, \lambda = 650nm$ For $n = 3, \lambda = 434.4nm$

For $\lambda = 650nm, 434.4nm$, there will be a maxima at 0

33 & 34.

Intensity of radiations at a distance 'l' from the point source $I = \frac{P}{4\pi l^2}$ where $P = 4\pi r^2\sigma T^4$

Power falling on the hole of the cavity $= \frac{I\pi d^2}{4} = \frac{P}{4\pi l^2}\frac{\pi d^2}{4} = \frac{Pd^2}{16l^2} = \frac{\pi d^2 r^2 \sigma T^4}{4l^2}$

Let the temperature of the cavity be T^{11}, the amount of radiation falling from inside the cavity on the hole will be $\frac{1}{4}\pi d^2 r\sigma(T'')^4$

Equating the two powers we get, $\frac{\pi d^2 r^2 \sigma T^4}{4l^2} = \frac{1}{4}\pi d^2\sigma(T'')^4 \Rightarrow T'' = T\sqrt{\frac{r}{l}}$

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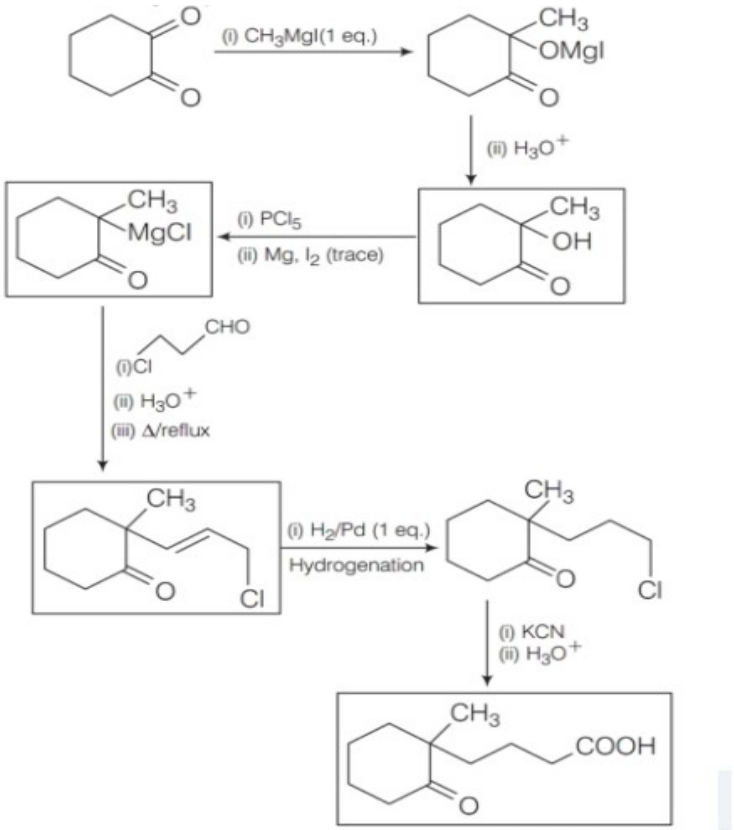
CHEMISTRY

35. Nucleophilic Substitution
36. Na/C2H5OH can not reduce acid.
37. Except statements (i) and (iv), all other statements are correct.
38. Picric acid is stronger acid than formic acid
39. B) NaOH is the limiting reagent if there is no formation of Na_2CO_3 (meaning only $NaHCO_3$ forms): This is incorrect. In this scenario, CO_2 would be the limiting reagent and NaOH would be in excess.
C) Equal masses of Na_2CO_3 and $NaHCO_3$ are formed if none of the reactant is left: This is incorrect. The amounts of products formed depend on the specific conditions (concentrations, etc.) and will not necessarily result in equal masses, or even total consumption of both reactants simultaneously to form only these two products. For both reactants to be fully consumed, a specific ratio of products would need to form, which can be calculated but does not necessarily result in equal masses.
40. These cations are of II or IV group as they are identified as sulphies.
Ksp of sulphide of group II cation < Ksp of sulphide of group IV cation. → Br– gives yellow ppt. with AgNO3,
which is partially soluble in NH4OH .



Brown ring test is for NO_2^- and NO_3^- So anions of salts of may be nitrite, nitrate and bromide

41.



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42.

$$Co^{2+} + 6NH_3 \rightleftharpoons [Co(NH_3)_6]^{2+} \qquad K_1 =$$

$$Co^{3+} + 6NH_3 \rightleftharpoons [Co(NH_3)_6]^{3+} \qquad K_2 =$$

$$Co^{3+} + e \longrightarrow Co^{2+} \qquad E^\circ = 1.9\ V$$

$$[Co(NH_3)_6]^{2+} \longrightarrow [Co(NH_3)_6]^{3+} + e \qquad E^\circ = -0.1\ V$$

$$Co^{3+} + [Co(NH_3)_6]^{2+} \rightleftharpoons Co^{2+} + [Co(NH_3)_6]^{3+}$$

$$E^\circ = \frac{0.06}{1}\log\frac{K_2}{K_1} \text{ (at equilibrium)}$$

$$1.8 = 0.06\log\frac{K_2}{10^5}$$

43. Water will vaporize till $P_{H_2O} = 0.04\,\text{atm}$

Now, $PV = nRT \Rightarrow 0.04 \times \left(40 \times 10^3\right) = \frac{w}{18} \times 0.08 \times 300$

Or $w = 1200 = 1.2kg$

$\therefore$ Percentage of water vaporized $= \frac{1.2}{5} \times 100 \qquad = 24\%$

44. $K_3\left[Fe(CN)_6\right]: Fe^{3+}: 3d^5$

$[FeCl_4]^{2-}: Fe^{2+}: 3d^5$

$\left[Fe(H_2O)_6\right]Cl_3: Fe^{3+}: 3d^5$

$[FeF_6]^{4+}: Fe^{2+}: 3d^5$

$\therefore p + Q + R + S = 1 + 3 + 3 + 2 = 9$

45. No of spectral lines given by H $= \frac{5 \times 4}{2} = 10$

No of spectral lines given by $He^+ = \frac{4 \times 3}{2} = 6$

No of spectral lines given by $Li^{2+} = \frac{3 \times 2}{2} = 3$

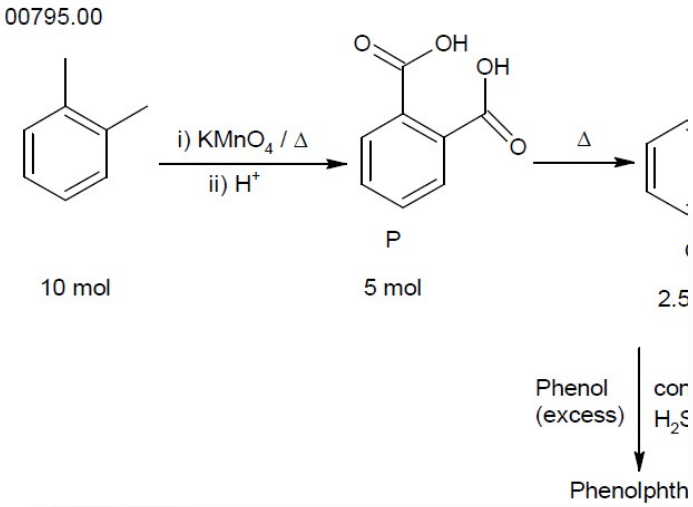
However

$\lambda_{2\to1}(H) = \lambda_{4\to2}(He^{2+})$

Total no of spectral lines = 10 + 6 + 3 – 1 = 18

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46.



47.

A = T 2 H-bond

G ≡ C 3 H-bond

Number of A=T pair = 7

Number of G≡C pair = 6

Number of H-bond involve in A = T = 7

Number of H-bond involve in G ≡ C = 6

Energy required for A = T = 14 × 1 = 14

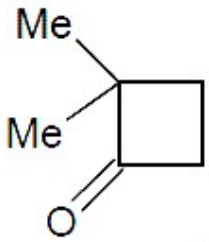
48 & 49.

$$2BF_3 + 6NaH \xrightarrow{450K} \underset{(P)}{B_2H_6\,(g)}$$

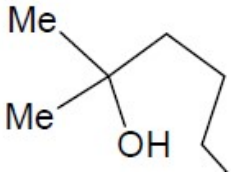
$$\underset{(P)}{B_2H_6} + 6H_2O \longrightarrow \underset{(R)}{2H_3BO_3} + 6$$

$$B_2H_6 + 3O_2 \longrightarrow B_2O_3 + 3H_2O$$

50 & 51.



. compound B



COMPOUND D

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